Stat 6410 (Ghosh) Midterm 2

Write legibly showing all the steps VERY clearly. (Total points = 110, ***Max Score is =100.****)*

Let $\Omega \neq \emptyset$, where $\emptyset$ = empty set and $\mathcal{P}(\Omega)$ is the collection of all subsets of Ω, i.e the power set. Let $\mathcal{G}, \mathcal{F} \subset \mathcal{P}(\Omega)$ be a classes of subsets of Ω, and λ, μ be set functions.

1. Let m be the Lebesgue measure on $[0,1]$ and $\mathbf{1}_A(x)$ denote the indicator function of A. Define a sequence of measurable transformations on $(\Omega = [0,1], \mathcal{F} = \mathcal{B}[0,1], \mu = m)$: For $n \geq 1$, and $x \in \Omega = [0,1]$,

$$f_n(x) = n\mathbf{1}_{\left[\frac{1}{n}, \frac{2}{n}\right]}(x) = \begin{cases} n & \text{if } \frac{1}{n} \leq x \leq \frac{2}{n} \\ 0 & \text{otherwise} \end{cases}$$

The limit function is $f(x) \equiv 0$ for all $x \in \Omega = [0,1]$.

$(10 + 10 + 10 + 10 + 10 = 50$ points$)$

   (a) Show that $f_n \to f$ almost everywhere (m).

   (b) Does the sequence $f_n \to f$ in measure?

   (c) For what values of $0 < p < \infty$ does the sequence $f_n \to f$ in L_p?

   (d) Is this sequence $\{f_n\}$ uniformly integrable?

   (e) Consider the sequence $\{g_n\}$ where $g_n = \cos(f_n)$ (here $\cos(x) = \text{cosine}(x)$ is the well-known trigonometric function). Does this sequence $\{g_n\}$ converge to $g \equiv 1$ almost surely (m)?

2. (i) Show that a Binomial$(n = 10, p = 0.1)$ random variable exists:
   In other words, using steps of the Caratheodory extension theorem, exhibit a probability space $(\Omega, \mathcal{F}, P)$ and a random variable $X : \Omega \to \mathbb{R}$, such that its p.m.f. is:

$$P(X = k) = f(k) = \binom{10}{k}(0.1)^k(0.9)^{10-k}, \quad k \in \{0, 1, \ldots, 10\},$$

   and $P(X = k) = f(k) = 0$, otherwise.

   (ii) Compute the following Lebesgue integrals (give a numerical answer)

$$\int_\Omega X^2(\omega)\, dP(\omega)$$

You can state & use any pre-measure theory facts you know (if needed), but justify all the measure theory steps.

$(10 + 15 = 25$ points$)$

3. Answer the following:

$(15 + 10 + 10 = 35$ points)

(a) For a *nonnegative* random variable X, prove $\sqrt{1 + (EX)^2} \leq E\sqrt{1 + X^2} \leq 1 + EX$, considering the possibility that $E(X)$ can be infinite.

(b) Let $(\Omega = \mathbb{R}, \mathcal{F} = \mathcal{B}(\mathbb{R}))$. Let m be the Lebesgue measure and μ be the standard normal (Gaussian) measure. That is,

$$\mu(A) = \int_A \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\, dm(x), \quad A \in \mathcal{B}(\mathbb{R})$$

Show that the RN derivative of Lebesgue measure m w.r.t. μ to be:

$$\left(\frac{dm}{d\mu}\right)(x) = \sqrt{2\pi}\, e^{\frac{x^2}{2}}, \quad x \in \mathbb{R}.$$

(you can state any result proved in class to argue this)

(c) Let $(\Omega, \mathcal{F})$ be a measurable space with two measures μ and ν on it. Given,

$$\mu \ll \nu, \quad \text{and} \quad \mu \perp \nu.$$

Show that $\mu(A) = 0$ for all $A \in \mathcal{F}$.